

Verification II_Rate_and_State_Griffith

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1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator Verification II_Rate_and_State_Griffith correctly reproduces the results of Garagash (2021). This is done by digitizing the data of Figures 5 and S5 in Garagash (2021) and comparing them to the simulation results produced by the MATLAB and Python simulators.

2 Results

The results for the aging and slip law simulations can be found in Figures 1 and 2, respectively.

3 Conclusion

The simulators have both been verified.

References

D. Garagash. Fracture mechanics of rate-and-state faults and fluid injection induced slip. *Philosophical Transactions of the Royal Society A*, 379, 2021. doi: <https://doi.org/10.1098/rsta.2020.0129>.

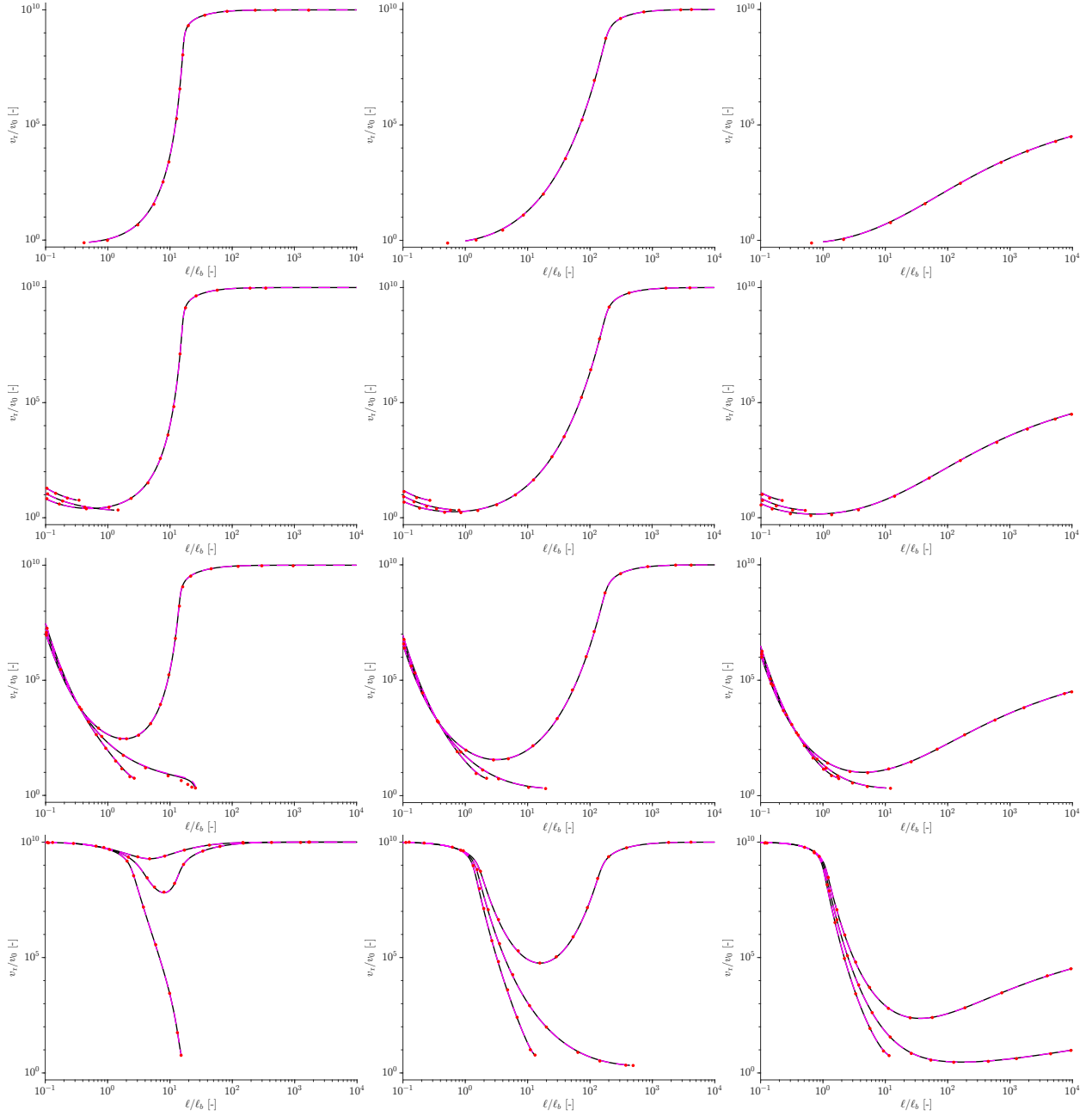


Figure 1: Rupture velocity as function of crack length for MATLAB (black) and Python (dashed pink). Each line corresponds to a different background stress. The digitized results of Garagash (2021) are shown with red dots. This figure corresponds to the aging-law simulations of Figure S5 of Garagash (2021), with a/b and ΔT in the same order as Figure S5 of Garagash (2021).

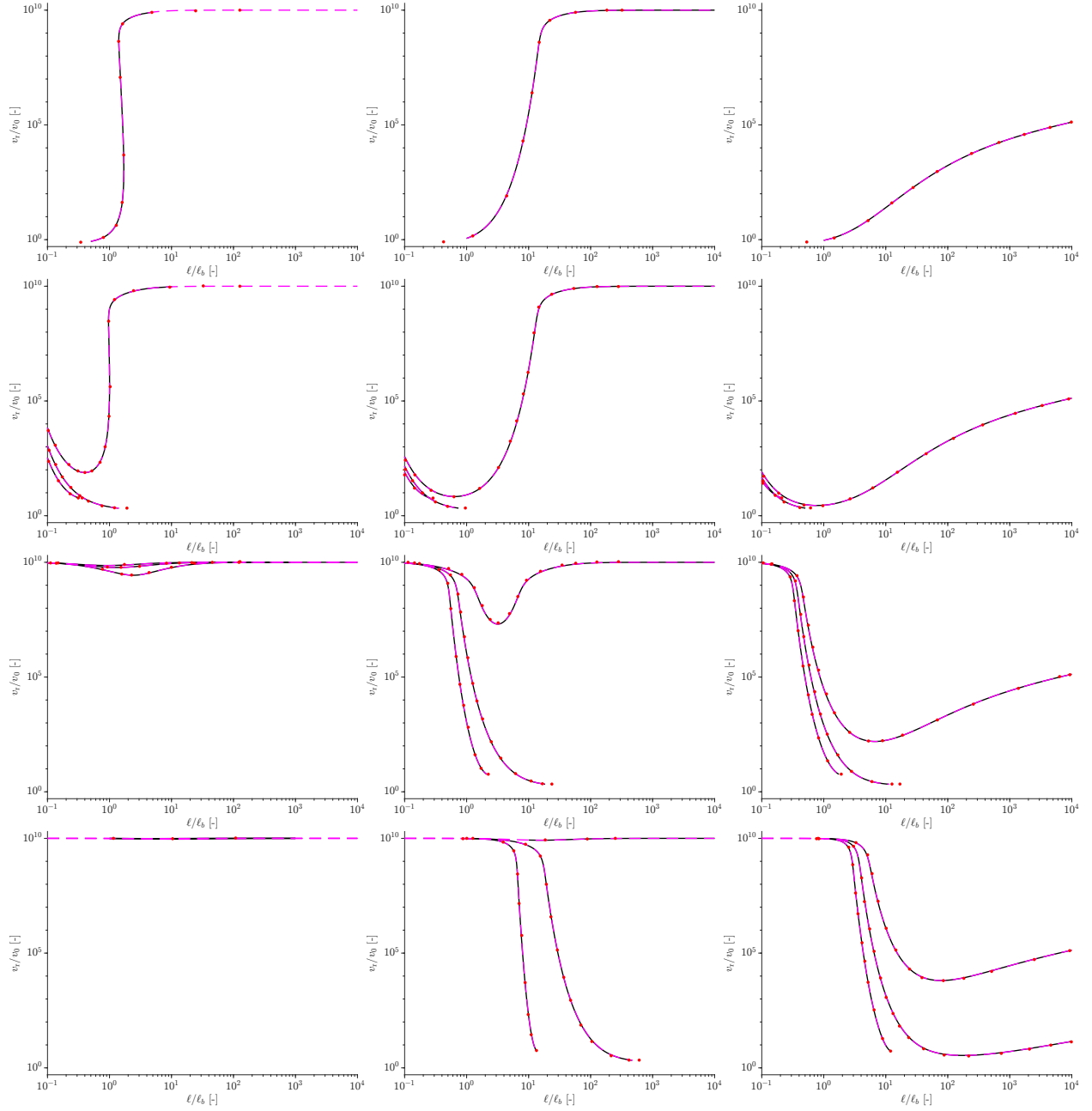


Figure 2: Rupture velocity as function of crack length for MATLAB (black) and Python (dashed pink). Each line corresponds to a different background stress. The digitized results of Garagash (2021) are shown with red dots. This figure corresponds to the slip-law simulations of Figure 5 of Garagash (2021), with a/b and ΔT in the same order as Figure 5 of Garagash (2021).